

Problem Set 1: Randomized Experiments

Instructions: To answer the questions in this problem set, please do not just attach the program output and expect me to find the answers there. **Answers should be clearly stated or will receive zero points. Preferably, turn in your answers separately from the program output.** In any case, all the code and the program output should be attached as an appendix to your problem set. **Problem sets without code will receive a score of zero. Also, no late submission of the problem set will be accepted.**

Setting: You recently got an internship at the Manpower Demonstration Research Corporation (MDRC), which is a “nonprofit, nonpartisan education and social policy research organization dedicated to learning what works to improve programs and policies that affect the poor,” as stated in its website.¹ Jerzy Rubin, your supervisor and an expert on causal inference, always complains to his colleague Donald Neyman that the only thing his economics and other “analytics” interns know how to do is run regressions (or machine learning methods), without ever thinking hard about what they are really doing or if the effects they estimate are really causal. To prove his point, he typically gives his interns one of their classical social randomized experiments on evaluation of training programs to see what they do with it (“besides running regressions”, he usually says). He does the same with you this time, and asks you to write a one-page report on your findings. Of course, he does not know you took ECON 440 at Cal Poly ☺. For the purpose of this problem set, **do not** write the one-page report (I’ll probably ask that next year ☺). The questions below would be part of the main analysis you would perform for such report.

Problem: You are asked to evaluate the effects on earnings of the National Supported Work Demonstration (NSW) job training program, whose main objective was to help disadvantaged workers improve their labor market prospects by providing them with work experience and counseling.² During the mid-1970s, the MDRC performed a study on the effects of the NSW program. In it, individuals were randomly assigned to either receive or not job training. Baseline earnings were measured in 1975, which was the year before they were assigned to either the treatment or control group. Earnings were subsequently measured in 1978, a few years after the treated individuals had gone through the NSW program (the program was about 12 to 18 months long). The program focused on four groups: women on Aid to Families with Dependent Children (AFDC) for many years, ex-criminal offenders, ex-drug addicts, and high school dropout (for the last three groups, both men and women were considered).

Data: The data set you will use is called `nsw_econ440.dta` (available on Canvas). This data set, comprised of only males, includes the treatment variable `treat` (1 if treated, 0 if not treated), the outcome variable `re78` (real annual earnings in 1978), as well as the following pre-treatment

¹ To learn more about it, go to <http://www.mdrc.org>, and hit the tab “About MDRC”.

² You can learn more about the program here: <http://www.mdrc.org/publication/summary-and-findings-national-supported-work-demonstration>.

variables: *age* (in years), *education* (years of education), *black* (1 if black, 0 otherwise), *hispanic* (1 if Hispanic, 0 otherwise), *married* (1 if married, 0 otherwise), *nodegree* (1 if no high school degree, 0 otherwise), and *re75* (real annual earnings in 1975).

Questions:

1. For a given individual i , clearly define what the potential outcomes $Y_i(1)$ and $Y_i(0)$ would be in the context of this problem.
2. First, provide evidence that the randomization worked in balancing the characteristics of the treatment and control groups. To do this, for all pre-treatment variables calculate the average in the treatment and control groups and test whether the difference is statistically significant. Please create a *clean* table which includes the following columns: treatment group mean, control group mean, difference, and its corresponding t-statistic or p-value (i.e., a table like the ones discussed in class). In addition, next to each difference put in superscript: one asterisk if the difference is statistically different from zero (i.e., “statistically significant”) at the 10% significance level, two asterisks if statistically significant at the 5% significance level, or three asterisks if statistically significant at the 1% significance level. The table should be comprehensive on its own, in the sense that it should be clear what the table is showing (e.g., it should have a title, and a note at the bottom indicating what the asterisks mean, etc.). Again, suppose you will be providing this table to your supervisor Dr. Rubin.
3. Based on the previous table, do you think the randomization worked in balancing the treatment and control characteristics? Why or why not? What do these results suggest about differences in the *unobserved* characteristics of treatment and control groups?
4. Estimate the average treatment effect (*ATE*) assuming the randomization was successful (remember to also provide its standard error). In addition, answer the following questions: (a) how do you interpret that number? (b) Is the *ATE* statistically different from zero? At what significance level (e.g., 1%, 5%, 10%)? (c) Is this effect “economically significant”? In other words, is the effect large or small in a practical sense (say, would it make a difference for these individuals)?
5. Now you will estimate linear regression models using all pre-treatment variables as controls in order to increase the precision of your estimate of the *ATE* and adjust for any remaining imbalance in the covariates that may have occurred despite the randomization.³ You will run four linear regressions, with each adding more regressors. The regressor in the first regression is only the treatment variable; the second regression adds the demographic variables (*age*, *black*, *hispanic*, *married*); the third adds to the previous one the education variables (*education*, *nodegree*); and the last one adds to the previous one the individuals’ labor market histories (*re75*). Please create a *clean* table showing your results. Each column of the table should correspond to a regression, and each row should show the regression results for each regressor and the constant term (both estimated coefficient and its corresponding standard error). At the bottom of the table list the number of observations and R^2 for each regression. Finally, next to each estimated coefficient put in superscript: one asterisk if it is statistically different from zero (i.e.,

³ Remember that, due to the randomization, any imbalance in covariates would likely be small, so linear regression usually works well in this context to adjust for them (remember that linear approximations usually work well for small changes or differences; for example, think about calculus—derivatives, differentials, Taylor expansions, etc.).

“statistically significant”) at the 10% significance level, two asterisks if statistically significant at the 5% significance level, or three asterisks if statistically significant at the 1% significance level. Like in question 2, the table should be comprehensive on its own, in the sense that it should be clear what the table is showing (e.g., it should have a title, and a note at the bottom indicating what the asterisks mean). You may want to see the file “Sample_Table_for_Question_5_PS1.pdf” on Canvas. Again, suppose you will be providing this table to your supervisor Dr. Rubin.

6. Based on the previous table, (a) what is the effect of adding covariates to the regression of the outcome on the treatment variable on your *ATE* estimate from question 4? (b) What do you think this table suggests would happen if we were able to add to these regressions *unobserved* variables (i.e., variables that we do not have in our data set), such as motivation, ability and interpersonal skills?
7. Based on your analysis in all the above questions, would you give a *causal* interpretation to the estimated effect in question 4? In other words, do you think the estimated effect in question 4 reflects the *causal* effect of the NSW program on the individuals’ earnings? Why or why not?
8. Your new friend, who is also an intern at the MDRC but did not take ECON 440 at Cal Poly, tells you that it would be good to estimate the average treatment effect on the treated (*ATT*) which, from what he/she heard at a recent meeting, gives the average effect of the program on the earnings of those individuals who participated in the program (instead of the effect for the whole target population). You just kindly smile at him/her when he/she tells you this. Provide your estimate of the *ATT*, along with its standard error and a *brief* explanation on how you got your result.
9. Your friend, who thinks it would be really good to impress Dr. Rubin, tells you: “I know what we can do. A big topic in economics is the causal effect of education on earnings. Since we have a randomized experiment, we can analyze the *causal* effect of not having a high school degree (*nodegree*) on earnings (*re78*). Let’s estimate this causal effect by comparing the average 1978 earnings of those who do not have a high school degree to the average 1978 earnings of those who have a degree. Let’s do that!” Would you agree with him/her? Why or why not? (Note: doing some of the analysis you did in questions 2 and 5 above would be helpful when discussing this with your friend. You do not necessarily have to create those tables again for this case to get full credit for this question, but I believe doing so will help you understand the material better, and it may help you when writing your answer. Of course, please feel free to turn in those tables, but do not feel you need to do them to get full credit.)